

Emmanuel Alabi

Software Engineer working toward AI & Robotics Research

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Research Interests

Embodied AI; robot learning; computer vision; autonomous systems; multimodal intelligence. Specifically, how an agent's representation of a physical environment should be shaped by having to move through it, and when learned components genuinely outperform well-specified classical methods.

Broader interests: machine learning, robotics, state estimation, planning, reinforcement learning.

Education

Babcock University — BSc (Hons) Software Engineering 2019 — 2023

Ilishan-Remo, Ogun State, Nigeria. School of Computing and Engineering Sciences.

Relevant coursework: Algorithms and Data Structures; Discrete Mathematics; Introductory Statistics; Introduction to Operations Research; Artificial Intelligence and Applications; Introduction to Big Data Engineering; Computer Organization and Assembly Language; Operating System I; Database System Design, Implementation and Management.

Research & Technical Experience

Risk-Calibrated Semantic Navigation Under Distribution Shift 2026

Preregistered ROS 2 / Gazebo benchmark. github.com/alabiemmanuel177/risk-calibrated-semantic-navigation

- Designed and executed a preregistered simulation study testing whether calibrated semantic perception uncertainty, converted into Nav2 navigation cost, improves closed-loop robot outcomes under controlled distribution shift; four systems share identical maps, routes, planner and controller families, varying only the treatment component.
- Hash-locked the protocol, manifest, models, thresholds, exclusions and analysis code before protected execution; the confirmatory campaign produced 6,840 unique schema-valid episodes, 6,820 entering analysis.
- Calibration (H1) and clean-route efficiency noninferiority (H3) passed; the preregistered collision-reduction (H2) and severity-interaction (H4) hypotheses did not. Better component calibration did not establish safer navigation — reported as the result, with integrity deviations and limitations disclosed.
- Independent clean-clone reproduction regenerated all hypothesis JSON and key tables byte-for-byte (249 tests). emmanuelalabi.com/research/risk-calibrated-semantic-navigation

Independent Research Programme — Embodied & Multimodal Autonomy 2026 — present

- Self-directed programme building the mathematical, machine-learning and robotics foundations required to conduct independent research: linear algebra, probability, optimisation, state estimation and rigid-body geometry, each with implementations written from first principles.
- Reproducing published baselines in perception and navigation before extending them, on the principle that a reproduction which fails is itself a result worth documenting.
- Working toward a research question in embodied, multimodal autonomy precise enough to investigate with controlled experiments rather than demonstrations.
- Programme and progress documented publicly at emmanuelalabi.com/research/research-programme-2026

Independent Research & Reproduction Studies

Driver Drowsiness Detection — Reproduction & Evaluation Study 2026

Independent computer vision study. github.com/alabiemmanuel177/driver-drowsiness-reproduction

- Rebuilt and evaluated a classical eye-state and temporal-alerting pipeline under a subject-disjoint protocol on the MRL Eye Dataset (84,898 images, 37 subjects), training a lightweight CNN from scratch against a HOG+SVM baseline and measuring frame- and event-level behaviour.
- Designed an oracle-classifier decomposition showing that once classification accuracy was sufficient, the historical temporal threshold — not residual classification error — remained the dominant source of false alerts under a simulated-sequence protocol.
- Built a fully reproducible pipeline: deterministic splits, automated leakage checks, three-seed ablations, generated result artifacts, structured failure analysis, and 145 automated tests.
- emmanuelalabi.com/projects/driver-drowsiness-reproduction

Selected Academic Projects

- Co-designed and implemented a web-based learning management system for Babcock University's Centre for Open and Distance e-Learning, supporting student, lecturer and administrator workflows.
- Built course and resource management, assignments, authentication, user administration, collaborative course activity and role-specific interfaces using React, Node.js/Express and MongoDB.
- Conducted requirements analysis, system design, iterative implementation under a spiral methodology, and interface/usability, unit and integration testing.
- emmanuelalabi.com/projects/bucodel-learning-management-system

Selected Engineering Projects

Obelo — Multimodal AI Infrastructure

- Production AI platform combining multimodal generation, retrieval-augmented context, asynchronous orchestration, model routing and evaluation systems.
- Experimentation across multiple image-generation and language-model families; structured creative pipelines; reliability mechanisms for long-running AI workflows.
- emmanuelalabi.com/projects/obelo-multimodal-ai-infrastructure

Selected Technical Experience

Founder / Lead Engineer — Obelo

- Designed and developed a production AI platform: multimodal generation, retrieval-augmented brand context, model routing across model families, asynchronous orchestration, and evaluation and quality systems.

Software Engineer — SPay Business

- Production mobile-financial systems: authentication improvements, application performance, interface redesign, CI/CD automation and over-the-air mobile deployments.

Technical Skills

Programming Python, TypeScript, JavaScript, Java, SQL

AI/ML Computer vision, multimodal AI systems, retrieval-augmented generation, LLM and model integration, evaluation systems

Systems & engineering React, Next.js, Node.js, Express, MongoDB, PyTorch, PostgreSQL, Redis, Docker, CI/CD